

Gurpreet Singh

DEVOPS ENGINEER

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Profile Summary

A results-driven DevOps Engineer with extensive expertise in designing and optimizing CI/CD pipelines, container orchestration, and scalable cloud infrastructures. With a solid technical background and proven experience in automation, and excel at collaborating with cross-functional teams to deliver secure and high-performance solutions that drive business growth.

Experience

TestingXperts | DevOps Engineer (Jan 2022 – Current)

- Automated **AWS** infrastructure provisioning using Terraform and Jenkins for seamless CI/CD pipelines, enhancing development efficiency and infrastructure reliability.
- Optimized AWS VM instance costs by implementing an AWS Lambda function to automatically shut down unused instances during specific time periods.
- Developed and deployed Python applications on **Google Cloud Platform**, utilizing Cloud Build for CI/CD pipelines and deploying the applications on GCP
- Spearheaded the migration of repositories from Git and **SVN** to GitHub using **Ansible**, Jenkins, and custom shell scripts, comprehensive verification and updating Jenkins jobs for streamlined workflows.
- Designed and implemented Azure DevOps pipelines for Terraform-based Azure infrastructure creation, optimizing pipeline efficiency with conditional logic.

Projects

Deluxe Corporation (DPP)

- Designed and implemented **Infrastructure-as-Code (IaC)** using Terraform & Terragrunt to provision **v-Sphere** infrastructure with reusable modules, automated triggers, and per-environment state management.
- Configured **backend storage in JFrog Artifactory** with separate repositories for development and production artifacts.
- Automated infrastructure provisioning and application deployment through **Harness pipelines**, managing both Terraform/Terragrunt workflows and CI/CD processes.
- Built CI pipelines for **13 enterprise applications** (.NET & Node.js) integrating SonarQube for quality, Checkmarx CxOne for security scanning, and automated unit testing with enforced quality gates.
- Developed CD pipelines to deploy artifacts from JFrog to **Dev, QA, XUAT, Validation, and Prod** environments with Harness approvals, automated rollback mechanisms, and deployment notifications in Microsoft Teams.
- Orchestrated **zero-downtime blue-green deployments** across all environments, improving release reliability and reducing post-deployment incidents.
- Integrated Jira and ServiceNow workflows for production governance, approvals, and change management.
- Deployed **Dynatrace monitoring** for real-time application & infrastructure observability, enabling proactive alerts and performance optimization.

Key Achievements:

- Standardized CI/CD processes across all applications, increasing deployment frequency and reducing failure rates.

Net Number Migrations

- Successfully led the migration of multiple Git and SVN repositories to GitHub with zero disruption to development workflows.
- Developed **Python** and **Shell** automation scripts to verify post-migration integrity, validating repository contents, commit history, and branch consistency.
- Automated source–destination comparisons for commit count, file count, and checksum accuracy, ensuring complete and error-free transfers.
- Reconfigured **Jenkins** jobs to point to new GitHub URLs, updated **CI/CD** pipelines, and triggered clean builds to validate integration and deployment readiness.
- Implemented repository locking via **Ansible** to prevent commits to legacy Git and SVN repositories, maintaining integrity of the control of the integrity of the system.
- Collaborated with development teams to align migration schedules, eliminating merge conflicts and minimizing downtime.
- Recommended and initiated best practices for GitHub repository governance, including branch protections, PR reviews, and automated security scans.

SRA (Solicitors Regulation Authority)

- Maintained dedicated repositories for IaC and application code for clean separation of concerns and maintainability.
- Led Azure Landing Zone migration, provisioning Function Apps, Storage Accounts, Web Apps, and other resources via Terraform, with full lower-to-higher environment promotion workflows.
- Optimized cloud spend by upgrading WebApp SKUs (P2v2 → P3v0) for higher performance at reduced cost.
- Decommissioned legacy resources and pipelines post-migration, improving operational efficiency and reducing maintenance overhead.
- Configured Azure Front Door for global access layer, integrating with Reform, Identity, and Data Retention components for improved performance, security, and routing.
- Managed certificate and secret renewals via Azure Key Vault, zero downtime during updates and accurate mapping of secrets to components.
- Performed agent pool and VM updates to maintain Azure DevOps build compatibility and stability.
- Integrated SonarQube for static code analysis and Checkov for Terraform security/compliance enforcement in pipelines.
- Created and tracked user stories with linked tasks and releases, providing full visibility of migration progress and decommissioning activities.
- Developed Kusto Query Language (KQL) queries to build Azure dashboards for tracking application and infrastructure health, enabling proactive detection of anomalies and performance trends.
- Configured and managed Azure Alerts to trigger real-time notifications for error spikes and threshold breaches.
- Raised and tracked incidents via LittleFish (Service Request Tool), ensuring timely escalation to relevant teams and continuous updates throughout the incident lifecycle.
- Maintained detailed incident logs for audit compliance and trend analysis.
- Created and optimized Azure dashboards as per stakeholder requirements, aligning metrics with predefined thresholds.

Observability and Chaos Mesh

- Deployed Python and .NET applications on Amazon Elastic Kubernetes Service (EKS) using Jenkins pipelines.
- **AppDynamics** and Grafana for comprehensive application monitoring.
- Worked on a service mesh in applications and EKS using Chaos Mesh and set up visual graphs and alerts.
- Conducted research and development on auto-healing and auto-remediation for enhanced system resilience.

IAC

- Wrote Terraform scripts to create resources in both AWS and Azure.
- Worked with the development team to set up Terraform which provides output based on specific needs.
- Created Terraform scripts to check available cloud resources meet the required criteria.

Education

Skills

- Cloud Platforms: **AWS**, Azure, **GCP**
- Infrastructure Automation Tools: **Terraform**, Terragrunt
- Continuous Integration Systems: Jenkins, **Harness**, **AzureDevOps**
- Containerization Technologies: **Docker**
- Scripting Languages: BASH, **Python**
- Observability: Prometheus, **Grafana and prometheus**, AppDynamics, Dynatrace, **Loki and promtail**
- Security Automation Tools: SonarQube, Trivy, OWASP ZAP, **Burpsuite**, **Nessus**

Certifications

- **Harness** Continuous Integration, **Continuous Delivery** & GitOps Developer and Administrator
- **Google** Cloud Digital Leader